

Comparative Analysis of Dimensionality Reduction Methods for Image Classification

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Abstract

This research project examines how reducing image data affects the performance of classification machine learning models on the CIFAR-100 dataset. We compared three techniques for reducing data size including Principal Component Analysis (PCA), Uniform Manifold Approximation & Projection (UMAP), and Autoencoders to see how they influenced the accuracy and speed of different classifier architectures. Our experiments proved that using the original and unaltered images gives significantly better results than any of the simplified versions. The ResNet model using raw untampered data reached the highest accuracy at 65.82% whereas the best result using data reduced by PCA dropped to 28.69%. We also found that the UMAP method performed poorly across all tests with accuracy scores below 10%, suggesting that it lost the critical details needed to distinguish between different images. While PCA and Autoencoders resulted in lower average accuracy scores ranging from 26 to 29%, they were successful in making the model training process much faster compared to using the raw data.

1. Introduction & Objective

Processing high dimensional image data can have significant computational challenges for modern machine learning models. Dimensionality reduction offers a powerful solution to this issue by compressing feature vectors to retain essential information while discarding redundant data. However, selecting the right reduction technique is critical because the distinction between linear and nonlinear approaches impacts the final classification performance. Our research examines how specific methods including Principal Component Analysis (PCA), Uniform Manifold Approximation & Projection (UMAP), and Autoencoders affect both the accuracy and training speed of various neural networks. The main objective of this study is to observe which reduction method best preserves classification accuracy and to determine whether a basic Transformer architecture can outperform traditional Autoencoders in this domain. We further aim to establish which techniques provide the most interpretable feature representations and to find an optimal compression ratio that balances computational efficiency with classification accuracy. To answer this, we implemented an experimental design that evaluates 12 unique scenarios by combining four input types with three classifier architectures and measuring the resulting efficiency and predictive success.

2. Literature Review

Existing studies have explored the idea of feature compression and classification and often rely on both linear and non-linear dimensionality reduction techniques to reduce high-dimensional data, such as images [5]. Standard linear methods like PCA simplify data by identifying orthogonal directions of maximum variance, the principal components and projecting the data onto a lower-dimensional subspace, but it fundamentally fails to capture complex, non-linear relationships inherent in many real-world datasets. Manifold learning techniques such as UMAP [1] construct a high-dimensional graph of the data and optimize a low-dimensional equivalent, which preserves the complex topological and neighborhood structure which is crucial for downstream tasks. Additionally, deep learning approaches, specifically Autoencoders [6], give a robust alternative, learning an optimal, compressed latent space through a reconstruction objective, which serves as a highly effective feature extractor. When coupling these features with established classifier architectures, researchers often turn to Transfer Learning using models like ResNet-50 [2], utilizing the residual connections and foundational knowledge pre-trained on diverse datasets like ImageNet, or use recent innovations like the Transformer [4], which uses self-attention mechanisms to process image patches as sequences of tokens.

Building on prior work on high-dimensional image classification, we adopt a three part strategy designed to balance classification accuracy and computational efficiency. For raw untampered images, we use a frozen ResNet-50 backbone with transfer learning on the final classification layer [2], leveraging its well-established feature extraction capability while avoiding unnecessary retraining cost. After dimensionality reduction to 512-dimensional vectors where spatial structure is largely discarded, we replace convolutional models with a lightweight MLP trained from scratch, which is better suited to non-spatial embeddings. Finally, to explore whether joint feature learning and classification can outperform these separate pipelines, we introduce a hybrid autoencoder architecture that learns compressed representations and class predictions at the same time. The combined approach lets us compare the conventional pipelines with a more integrated model that learns representations and classifications together.

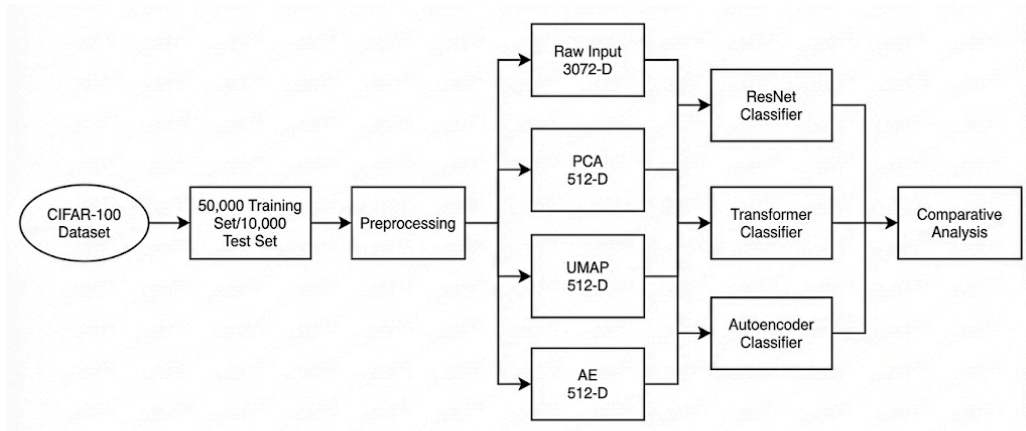
3. Methodology

Dataset: The CIFAR-100 dataset consists of 60,000 color images (32x32x3), distributed across 100 different classes. For our experiment we split the dataset into 50,000 images for the training set and 10,000 images for the test set. The images were flattened to 3,072-D and normalized to [0,1].

Dimensionality Reduction: The dimensionality reduction methods (PCA, UMAP, Autoencoder) were first fitted just on the training set (50,000 images). The learned transformation was then applied to both the training and the test sets to prevent data leakage. We chose to go with 512 dimensions for all reduction methods because PCA retains 95.23% of the total variance at this level.

Experimental Design: We designed and evaluated 12 experiments by combining four input types (Raw 3072-D, PCA 512-D, UMAP 512-D, Autoencoder 512-D) with three classifiers (ResNet, Transformer, Autoencoder). Each combination was trained and evaluated on classification accuracy and training time, as shown in **Figure 1**.

Figure 1: Full workflow from raw CIFAR-100 images to final analysis.



4. Results & Comparative Analysis

Table 1 summarizes the classification accuracy across the 12 experiments. ResNet with raw features achieved the highest accuracy at **65.82%**. As expected, all dimensionality reduction consistently decreased the performance. UMAP performed the worst (1-10% accuracy), while PCA and Autoencoder reduction had very similar results around **26%** average accuracy. The Raw features averaged **42.98%** accuracy across classifiers, which was a better average accuracy than all reduction methods.

Table 1: Classification Accuracy Results (%)

Model	Raw	PCA	UMAP	AE
ResNet	65.82%	28.69%	10.07%	28.75%
Transformer	35.45%	30.79%	1.00%	27.74%
Autoencoder	27.68%	20.22%	8.20%	20.57%
Average	42.98%	26.57%	6.42%	25.69%

ResNet Performance

ResNet had an accuracy of **65.82%** on raw images, which outperformed all other experiments. The pretrained features captured the important visual patterns like edges, textures, and shapes, which were transferred well to the CIFAR-100 dataset. The Transformer (trained from scratch) only reached **35.45%** on raw images, showing the importance of pretraining for image classification. When reduced to 512-D vectors, ResNet relied on a MLP classifier and lost accuracy. It dropped to around **29%** in accuracy in comparison to the other models that we trained from scratch.

PCA vs Autoencoder Reduction

The linear PCA reduction with an average accuracy of **26.57%** performed similarly to the nonlinear Autoencoder reduction, which had an average accuracy of **25.69%**. This tells us that for the CIFAR-100 images, linear relationships capture most discriminative information. The added complexity of the autoencoder did not give any meaningful benefits. This was most likely because it was trained only with reconstruction loss instead of using class labels that emphasize more discriminative features.

UMAP Performance

UMAP reduction had around only **1-10%** accuracy across all the classifiers, which seems like it was making random predictions for 100 classes. This might be because UMAP is optimized for 2-3D visualization not 512-D output, and its manifold assumptions might break down at high dimensions. Also, UMAP optimizes for neighborhood preservation instead of class separability, which means 2 images from different classes can be neighbors in the pixel space.

Efficiency vs Accuracy Trade-off

As shown below in **Figures 2-4**, dimensionality reduction reduced around **50-98%** training time but with a large cost for accuracy. For ResNet, **98%** computation time was saved but for **56%** of accuracy loss. For Transformer and Autoencoder classifiers, the accuracy loss is not as large, with only **5-15%** accuracy loss. This makes reduction a valid option when considering ways to reduce computation time.

Figure 2: Classification accuracy per model reductions

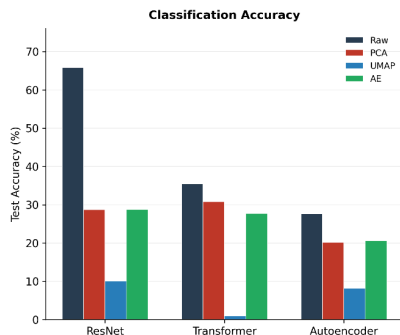


Figure 3: Training time per model accuracy changes & time reductions

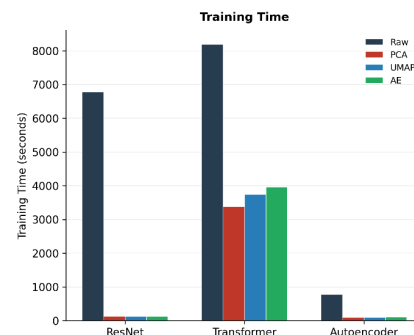
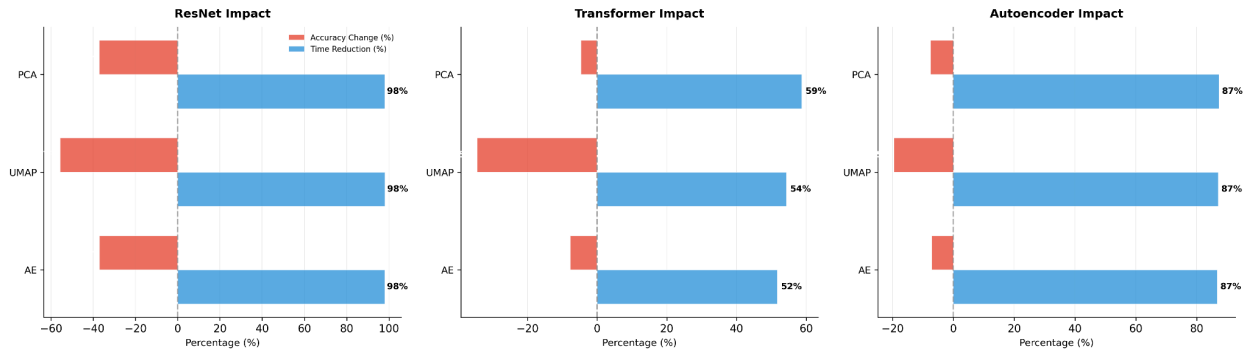


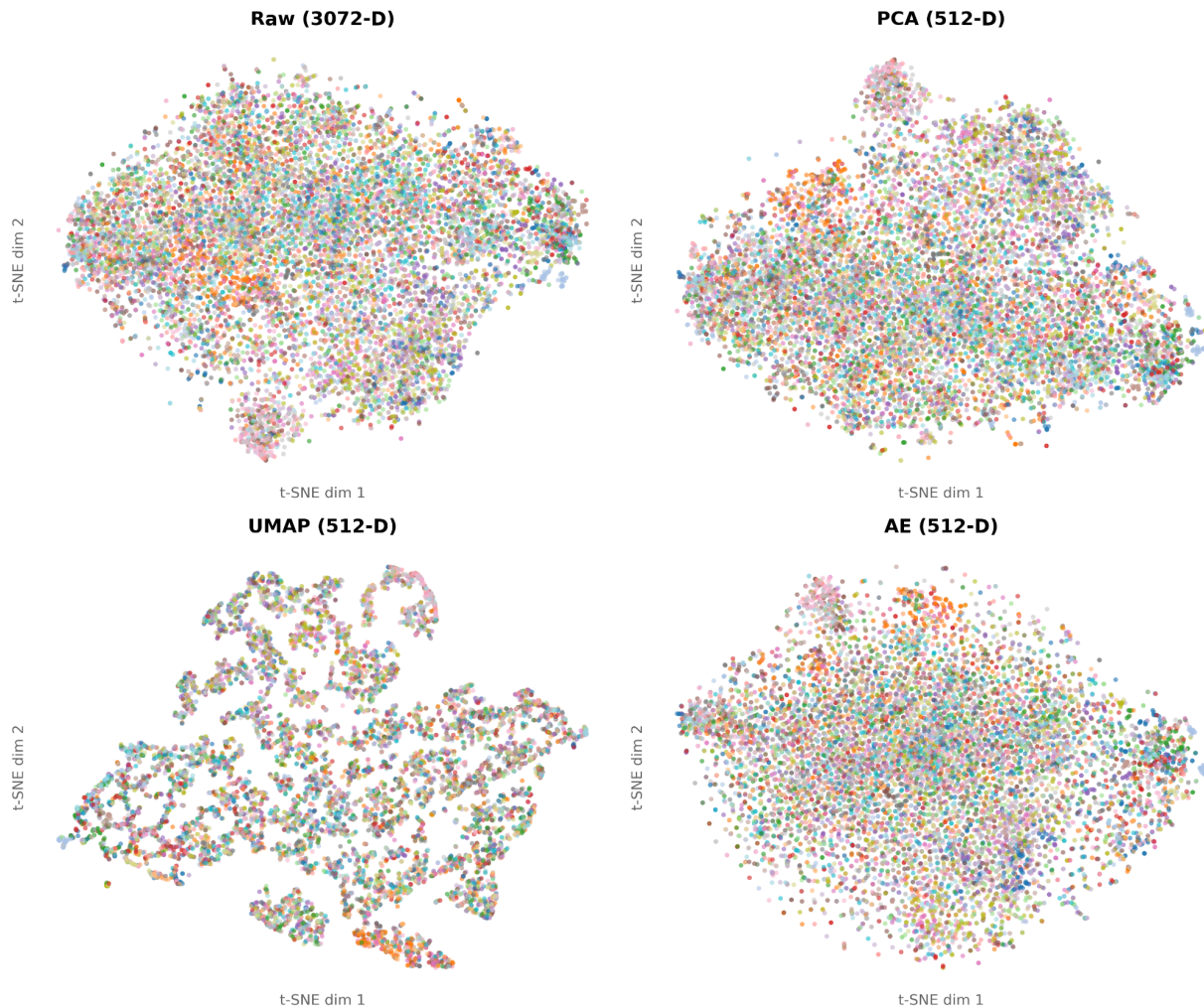
Figure 4: Accuracy for each classifier



Feature Space Visualization (t-SNE)

Figure 5 shows t-SNE projections of the four feature spaces. Raw and PCA features are spread out and overlap with some small clusters. The Autoencoder embedding looks similar, which makes sense since it also had a similar classification accuracy. UMAP has a broken ring-like clusters that do not match the class labels. This also helps show why UMAP performed poorly with its low accuracy.

Figure 5: t-SNE visualization comparing Raw, PCA, UMAP, Autoencoder feature spaces (for all classes)



5. Conclusion

In this study, we evaluated the trade-offs between classification accuracy, model complexity, and training efficiency on the CIFAR-100 dataset, revealing that the ResNet-50 model utilizing raw high-dimensional data resulted in the highest performance with a top accuracy of 65.82%. In contrast to this baseline, manifold learning via UMAP demonstrated a critical failure for feature extraction in high-dimensional spaces, giving accuracies below 10% and struggling to preserve global class-discriminative structures. Interestingly, we observed a similarity between linear and nonlinear reduction techniques, as the simple linear projection of PCA achieved 28.69% accuracy, performing comparably to the complex Autoencoder, which had an accuracy of 28.75%, which indicates that the computational overhead of training a reduction network gave no distinct advantage. Additionally, the Transformer significantly underperformed the ResNet model, highlighting the challenges of learning inductive biases without extensive pre-training.

To conclude, while dimensionality reduction suffers a considerable loss in accuracy, it does improve computation time (training speed). Specifically, PCA significantly reduced its training time, making it a practical choice for situations where computational resources are limited and speed matters more than accuracy.

6. Contributions of Team Members

- Sahil Dadhwal *implemented the experimental pipeline with data preprocessing, dimensionality reduction (PCA, UMAP, Autoencoder), all 3 classifiers (ResNet, Transformer, Autoencoder), training models & the visualization code.*
- Purva Khadke & Sarvesh Halbe *executed the initial code patching and exploration phase that identified limitations and necessitated the strategic pivot.*
- All team members *brainstormed the proposal, research roadmap and determined the comparative pathways for the experiments.*

References

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